

Program : Diploma in Textile Technology	
Course Code : 6062B	Course Title: High Performance Fibres
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To impart students with thorough knowledge on the manufacture, classification, properties & application of various high performance fibre

Course Prerequisites:

Topic	Course code	Course Title	Semester
Engineering Physics		Applied Physics I & II	1, 2
Engineering chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	U Explain the characteristics and applications of high performance fibre.	15	Understanding
CO2	Describe the characteristics, applications and production methods of PAN based Carbon Fibres	14	Understanding
CO3	Discuss the manufacturing process and properties of glass fibres, Lyocell and familiarize with the smart textiles	15	Understanding
CO4	Explain the manufacturing process, properties and applications of Poly lactic acid fibres, Bicomponent Fibres and nano fibres	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive levels
CO1	Explain the characteristics and applications of high performance fibre.		
M1.01	Discuss the general characteristics and applications of high performance fibre	4	Understanding
M1.02	Familiarize high strength high modulus fibres	4	Understanding
M1.03	List the different properties of Aramid fibres.	2	Understanding
M1.04	Explain the applications of Nomex and Kevlar fibres	5	Understanding
Contents: High performance fiber - General characteristics - Applications - classification - high strength high modulus fibers - Aramid - Properties - physical, chemical, thermal and mechanical - application of different aramid fibres - nomex, Kevlar.			
CO2	Describe the characteristics, applications and production methods of PAN based Carbon Fibres.		
M2.01	Describe the characteristics and applications of carbon fibre	3	Understanding
M2.02	Familiarize with the classification and properties of PAN based Carbon Fibres	3	Understanding
M2.03	Explain the Production process of carbon fibres from PAN precursors	4	Understanding
M2.04	Discuss the spinning methods adopted for spinning carbon fibers	4	Understanding
	Series Test - I	1	
Contents: Introduction to Carbon Fibres - General characteristics – Applications - PAN based Carbon Fibres - Classification based on Properties - Principles of Production of carbon fibres from PAN precursors – stabilization - Carbonization process - spinning methods - Centrifugal Spinning - Melt Blowing.			

CO3	Discuss the manufacturing process and properties of glass fibres, Lyocell and familiarize with the smart textiles.		
M3.01	Discuss the Glass Fibre, glass wool, optical fibres	5	Understanding
M3.02	Explain the manufacturing properties of lyocell fibres	4	Understanding
M3.03	Describe the smart textiles	3	Understanding
M3.04	Explain the characteristics of Absorbent and Superabsorbent Fibres	3	Understanding
Contents: Glass fibre - Manufacturing - Properties - Applications - Formation of Glass Wool - Optical Fibres - Lyocell Fibres - Manufacturing - Dope Preparation - Fibre Formation - Properties and Applications. - Compare Lyocell and Regenerated Viscose fibre - Smart Textiles - Shape changing Textiles - Temperature Responsive Textiles - Thermo-regulated textiles - Absorbent Fibres - Superabsorbent Fibres			
CO4	Explain the manufacturing process, properties and applications of Poly lactic acid fibres, Bicomponent Fibres and nano fibres.		
M4.01	Discuss the manufacturing process, properties and applications of Poly lactic acid (PLA) fibres	5	Understanding
M4.02	Explain the the manufacturing process and applications of Bicomponent Fibres	5	Understanding
M4.03	Familiarize with manufacturing techniques and applications of nano fibres	4	Understanding
	Series Test - II	1	
Contents: Poly (lactic acid) PLA fibres - PLA fibre Spinning - Solution - wet and dry Spinning – properties - Applications - Bicomponent Fibres - Spinning - Types - Core-Sheath Bicomponent Fibre - Side by Side Bicomponent Fibre - Segmented Pie - Applications - Nanofibres - Manufacturing Techniques - Electrospinning			

Text / Reference:

T/R	Book Title/Author
T1	High Performance fibres J W S Hearle.
R2	Textile fibres: Developments and Innovations, V. K. Kothari.
R3	Tao X (editor), Smart fibres, fabrics and clothing, Woodhead Publishing Ltd,Cambridge, 2001.
R4	D David R. Salem. (Eds). Structure Formation in Polymeric Fibres. Carl Hanser Verlag, Germany, 2001
R5	Physical properties of fibres J W S Hearle and W E Morton.

Online resources:

Sl.No	Website Link
1	http://nptel.ac.in/
2	www.textilelearner.blogspot.com
3	http://mytextilenotes.blogspot.in